Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

article [english]babel [utf8]inputenc johd

Title of your discussion paper

First Author Name[a*], Second Author Name[bc], etc.[ac]

[a]Department, University, City, Country

[b]Department, University, City, Country

[c]Department, University, City, Country

*Corresponding author: first name, initials, surname; email.address

Ringkasan

A short (up to 250 words) summary of the main contributions of the paper and the context of the research.

keyword 1; keyword 2; lower case except names, max 6

For determining author roles, please use following taxonomy: https://credit.niso.org/. Please list the roles for each author.

Discussion papers illustrate methods, challenges, and limitations in the creation, collection, management, access, processing, or analysis of data in humanities research, including standards and formats. These aspects must not necessarily be discussed with reference to a specific dataset (or collection thereof) but, if your paper focusses on particular datasets, we advise to add the dataset metadata under the section 'Dataset description'. Discussion papers have a word limit of 3,000-5,000 words, excluding title, author affiliations, abstract, Tables/Figures and references, but including Tables/Figures captions and footnotes. This template provides a general outline for discussion papers and authors can adapt the headings and include subheadings as they find appropriate. Please delete or replace the text not given in bold with your own text.

1 Context and motivation

Describe the context and motivation of your paper.

1.1 In-text citations

This journal uses a style based on the APA system (see here).
The following are some basic citation commands in LaTeX:

`\citet` → jensetmcgil
`\citet` → australiashealth
`\citet` → shree-a
`\citep` → fabricius-hansen2012b
`\citealp` → (eckhoff2018a)
`\citealp` → (eckhoff2018a; fabricius-hansen2012b; shree-a)

1.1.1 Other simple functions

To add bullet points:

- Some point
- Another point

Or numbered points:

1. Some numbered point
2. Another numbered point

This is an example of footnote[1].

This is a simple table:

Tabel 1: A caption.

1	2	3	4
a	b	c	d
e	f	g	h

Please refer to your table using: Table 1.

To add a figure, upload the figure into the `images` folder, and then embed it:



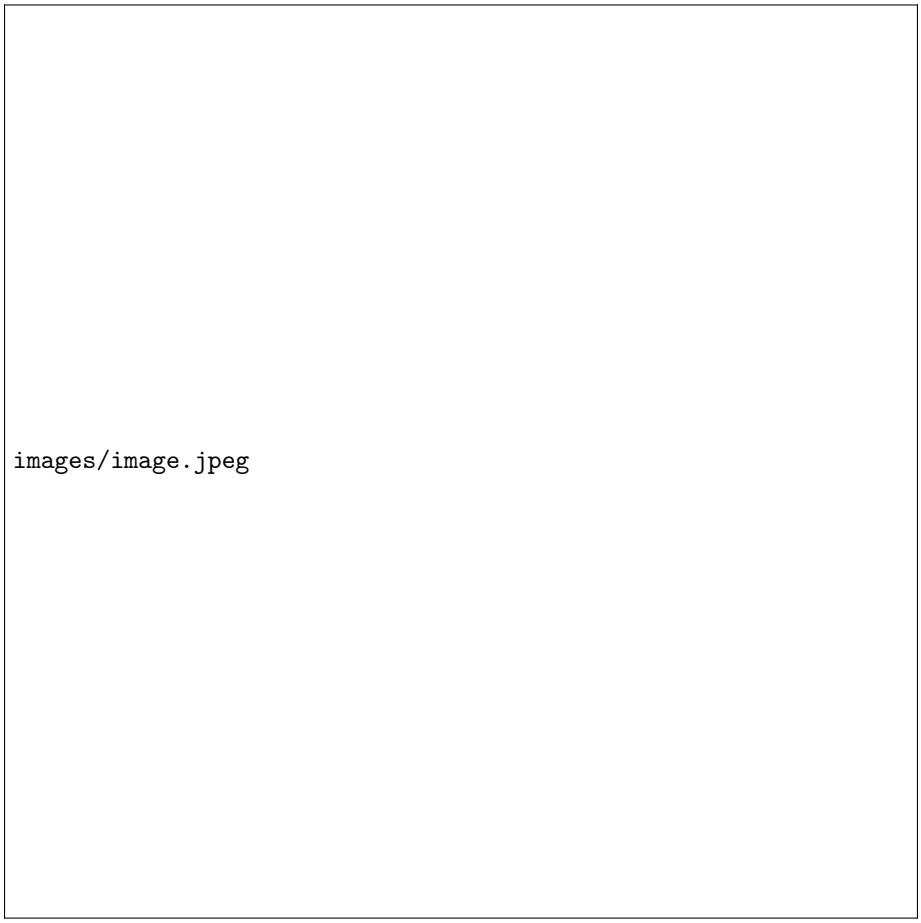
Gambar 1: JOHD's logo.

To resize the figure:



Gambar 2: JOHD's logo.

[1]This is a footnote



Gambar 3: JOHD's logo.

Please refer to your figures as: Figure 1, Figure 2, etc.

2 Dataset description

Here you can provide, if applicable, information about the dataset(s) whose creation, collection, management, access, processing or analysis have been discussed in this paper, following this schema:

Repository location A DOI of the openly available dataset being described is required.

Repository name The name of the repository to which the data is uploaded. E.g., Figshare, Dataverse, etc.

Object name Typically the name of the file or file set in the repository.

Format names and versions E.g., ASCII, CSV, Autocad, EPS, JPEG, Excel, SQL, etc.

Creation dates The start and end dates of when the data was created (YYYY-MM-DD).

Dataset creators Please list anyone who helped to create the dataset (who may or may not be an author of the data paper), including their roles (using and affiliations).

Language Languages used in the dataset (i.e., for variable names etc.).

License The open license under which the data has been deposited (e.g., CC0).

Publication date If already known, the date in which the dataset was published in the repository (YYYY-MM-DD).

3 Method

Describe the methods used in the study.

4 Results and discussion

Describe and discuss the results of the study.

5 Implications/Applications

Provide information about the implications of this research and/or how it can be applied.

Acknowledgements

Please add any relevant acknowledgements to anyone else that assisted with the project in which the data was created but did not work directly on the data itself.

Funding Statement

If the research resulted from funded research please list the funder and grant number here.

Competing interests

If any of the authors have any competing interests then these must be declared. If there are no competing interests to declare then the following statement should be present: The author(s) has/have no competing interests to declare.

Supplementary Files (optional)

Any supplementary/additional files that should link to the main publication must be listed, with a corresponding number, title and option description. Supplementary files should also be cited in the main text. Note: supplementary files will not be typeset so they must be provided in their final form. They will be assigned a DOI and linked to from the publication.